



DeepAID: a design of smart animal intrusion detection and classification using deep hybrid neural networks

S. Sajithra Varun¹ · G. Nagarajan²

Accepted: 17 April 2023 / Published online: 22 May 2023
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Abstract

Animal incursion is a significant problem in many locations, including roads, rural communities, and base camps for different camps. This has an influence on food security since the primer of animals will decrease agricultural land output and reduce farmers' profits. There must be a means to identify animals before they enter locations they should not be, so that appropriate measures may be taken to stop their incursion and its effects. Since a few decades ago, the study of detecting wild animals has been active. In this paper, a method for detecting animals is proposed utilizing developments in Deep Learning algorithm approaches. With the help of this deep learning technology, it is possible to identify animals in advance and take the proper security measures to prevent harm from animal entry. This study includes image processing and artificial intelligence for animal detection, species categorization, and automatic animal identification using CNN, an alarm unit, and an animal repellent circuit design. The proposed system experiments on the Amur Tiger Re-identification in the Wild datasets, Animals Detection Images Dataset, and Using Google Open Images V6+, the dataset for detecting animals (objects) was retrieved. Deep Convolutional Neural Network is used to extract these characteristics from cluttered photos. Hybrid CNN models were employed in the system's implementation, which was done on a regular camera trap database. Finally, CNN features were supplied to the top categorization deep learning algorithms. Our results show that our proposed method is better than other systems that have been mentioned in the literature.

Keywords Animal Intrusion · CNN · Hybrid CNN · Artificial Intelligence · Image processing · Animal repellent

1 Introduction

Animals are having a harder time surviving in their natural habitats as a conservative result of the rapid development in deforestation and the depletion of ecosystems. As a result, there have been countless occasions in which animals have escaped their enclosures in search of food. Animals virtually probably suffer injuries as an outcome of people acting voluntarily or because they are caught up in

risky human activities in these occurrences. As a result, we devised the technique of developing an image processing-based system that could distinguish the presence of animals in the area. As a result of the proliferation of security systems, it is not difficult to locate CCTV cameras in any given organization. Even if there are no cameras, it will not cost you too much to install one of your own. These optical technologies detect any attempt at sneaking in and keep your valuables secure as a result. On the other hand, in addition to capturing incursion, it may be intended to detect animal or human intrusion as well. When an animal is discovered using image processing, the security system will be capable of notifying the neighboring inhabitants so that they may take the needed safety measures. This type of real-time notice can assist prevent mishaps involving animals and also provide inhabitants with the information they need to hasten in order to resolve the problem.

In today's world, it is possible to collect massive amounts of information on the activities and behaviors of

✉ S. Sajithra Varun
sajithra_s@yahoo.co.in

G. Nagarajan
nagarajanme@yahoo.co.in

¹ Department of IIoT, MVJ College of Engineering,
Bengaluru, Karnataka, India

² Department of Computer Science and Engineering,
Sathyabama Institute of Science and Technology, Chennai,
India

wild animals across expansive space and extended periods of time. Because of their cheap costs and ease of operation, camera trap systems and many other modern devices may be utilized in the process of monitoring and analyzing animal populations. Because of the increase in available data on wildlife, research on wildlife-related topics, such as the impacts of climate change on wildlife, variations in habitat, the influence of human involvement on animals, and biodiversity across a variety of seasons, geographic areas, and species, has become easier to do. To create a stationary camera trap network for the purpose of monitoring local animals, sensor cameras are mounted on the branches of trees across an area. After activating the camera traps, each time motion is detected, a brief video of the animal's actions is captured together with information about its environment, like the light levels, humidity level, temperature, and position.

Deep neural networks are a collection of algorithms that have set new records in precision for various important tasks. A convolutional neural network (CNN) is a sort of deep neural network that is most commonly used for examining visual pictures. When compared to other image classification algorithms, CNNs utilize relatively little pre-processing on the images they analyze. One of the primary advantages of convolutional neural networks is that they do not require prior information or human involvement in the construction of their features (CNN). They have a number of applications in the fields of image and video recognition, recommendation systems, image classification, and medical image processing, among other areas. In the ground of ecosystem study, one of the most significant things to do is to investigate wild animals in the natural setting in which they live. Because of the enormous increase in the number of people living on Earth and the accompanying rise in the level of hunting for the persistence of economic development, there has been excessive exploitation of the planet's natural resources, which has led to rapid, inventive, and profound changes in the ecosystems of the planet.

Because of human activity, a growing portion of the surface of the land has been altered, which has the effect of altering the population of natural life through the way of well as their habitat and behavior. Even more tragically, many wild animals on Earth have vanished, and many species are being repositioned to other locations, where they can disrupt both natural and human resources.

This study intends to develop an IoT security system that detects motion in the farm environment, activates PiCam to take an image of the scene, and uses deep learning to identify the entity, such as an animal, person, or thing, that is producing the motion using MobileNets and Single shot detectors. On devices with limited resources, like smartphones and Raspberry Pi, these techniques can be combined and applied to identify objects in real-time and

very quickly. A pre-trained object detection network is loaded using OpenCV's Convolutional Neural Network (CNN) module (Sajith and Nagarajan 2022).

The essay is divided into five sections, with Sect. 2 providing a review of the relevant literature on the contributions made to security systems in smart agriculture employing IoT and image processing. Section 3 discusses the proposed system for smart agriculture, particularly to detect motion and malicious activities on farms and identify the animal causing motion. Section 4 presents the hardware and software components used in the framework and the image processing mechanism. Section 5 deals with experiments and result analyses and the task are concluded in Sect. 6.

2 Related works

The purpose of animal detection systems is to prevent accidents due to animal–vehicle collisions. This results in death, injury, and also property damage for humans.

Deep learning technology was utilized in this study by Yiting Li et al. (2018a) in order to achieve real-time and precise detection of surface flaws. Due to this, the only Shot MultiBox Detector (SSD) organize that was obtained was because of the metastructure, and it was combined with the bottom convolution neural system (CNN) MobileNet to form the MobileNet—SSD. It uses a parallel camera, an advanced camera, and a charge-coupled device (CCD) camera to gather target photographs, remove components, and put up a related scientific model, as well as bring an end to the MobileNetSSD. It relies, for the most part, on a paired camera, a computerized camera, a depth camera, and a charge-coupled device.

Trnovszky et al. (2017) investigated the performance of well-known image recognition algorithms such as Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA), Local Binary Patterns Histograms (LBPH), and Support Vector Machine (SVM) for the recognition rate of input animal photos. PCA, LBPH, LDA, and SVM's total recognition accuracy are shown. Next, animal recognition time is examined. All animal database experiments were run. This animal database has 500 topics (5 classes/100 photos). PCA features outperform LDA and LBPH for large training sets. LBPH outperforms PCA and LDA for short training datasets. 98% of planned CNN was recognized.

Kamencay et al. (2016) examined and compared PCA, LDA, and LBPH for animal picture recognition. This paper presents a comparative evaluation of prominent picture recognition algorithms' strengths and shortcomings. Two experiments compare performance. First, we test the recognition accuracy of PCA, LDA, and LBPH. Second,

we examine the total time for animal recognition. All 300 algorithms were evaluated (60 images for each class). PCA features outperform LDA and LBPH for large training sets. LBPH is better for short training data sets than PCA and LDA.

Li et al. (2018b) introduced a new hybrid technique based on density peaks clustering and k closest neighbors to boost the accuracy rate that DPNN was used to train and kNN was used for classification. The suggested DPNN approach has superior accuracy to the support vector machine, and there are many more machine learning methods. Vinayakumar et al. (2019) suggested a hybrid intrusion detection system (Scale-Hybrid-IDS-Alert Network) based on a scalable framework on a hardware server to categorize unanticipated cyber-attacks, and monitor network, and host-level events. The framework is built on a deep learning model using the DNNs approach for evaluating massive data in real-time and optimum network parameters and network typologies for DNNs. In their proposal for class imbalance processing for the IDS data set, Zhang et al. (2020) combined the Synthetic Minority Over-Sampling Technique (SMOTE) with under-sampling for segmentation relying on the Gaussian Mixture Model (GMM). Utilizing the UNSW-NB15 and CICIDS2017 data sets, their unique method's benefit is confirmed. This model illustrates a practical approach to dealing with data imbalance in an intrusion detection system (Thyagarajan and Minu 2013).

The majority of the techniques used have been centered on improving accuracy and precision and removing false

alarms in light of the aforementioned literature. For the categorization of multiple class assaults in the current article, we adopted an ensemble technique to improve the accuracy rate of the intrusion detection system. In order to speed things up and choose better features, we also applied deep learning. In order to locate wild elephants that affect human life and agricultural lands, Sailesh et al. (2019) proposed a system that uses an intelligent image processing algorithm on the Raspberry Pi. In accordance with our technique, the steps involved in the process include motion detection in the video frame and the identification of the objects in the area descriptor as the local features that characterize the distinctive characteristics of the animal. The intrusion detection system that comes closest to the strategy outlined in this work in terms of design is that presented by Angadi and Katagall (2019). Convolutional Neural Networks (CNNs) for object detection are used in camera traps to capture images after motion is detected. Using the Single Shot Multi-box Detection model (Liu et al. 2016), objects are located in the image, and if any wild animals are discovered, an alarm message is given to the owner instructing them to take the necessary action. The Raspberry Pi computer, which is situated right next to the camera trap, handles all processing.

Fig. 1 Overview of proposed methodology

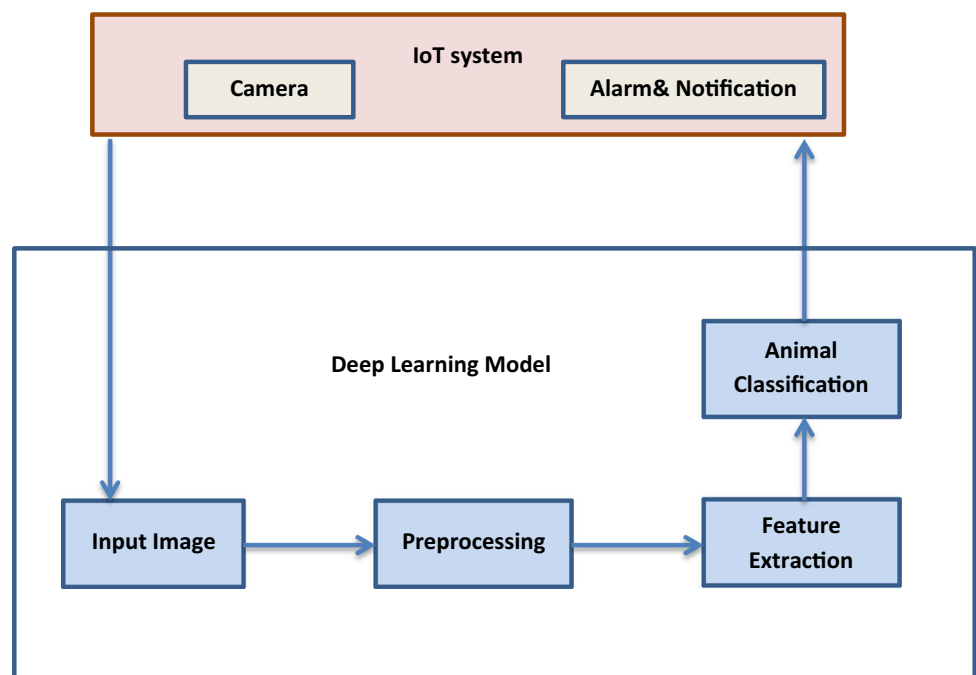
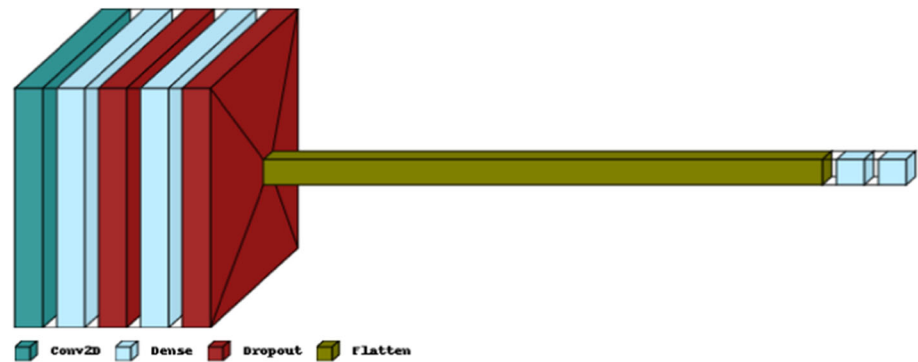


Fig. 2 Proposed model construction



3 Proposed method

Our creation of this smart solution, which employs a reliable mechanism to notify the owner in real-time that something has violated the perimeter, was motivated by the demand for intelligent solutions as shown in Fig. 1.

Additionally, the system makes customized noises for each type of trespass it detects in an effort to scare the intruder (for example, a dog's bark to frighten away livestock). This would warn the owner and allow them to take timely action, minimizing the harm brought on by the breach. The sound that is created is sufficient to preserve the crops on its own, even if the farm's owner was unable to visit due to some unforeseen circumstances. Convolutional Neural Networks, a type of Deep Learning technology, are extensively utilized by the system. By using a machine learning model or embedded digital image processing software in the microprocessor, the proposed

model recognizes not only motion but also the type of item (Fig. 2).

This classification knowledge would be sent into a program that interfaced with a speaker to generate a sound that was specifically tailored for each type of invader. With embedded systems, this may be improved. In that scenario, photos are captured using thermal cameras. After carefully evaluating a number of alternative sensors' day-and-night, range, and visualization capabilities, the decision to use a thermal camera was made. From an economic standpoint, a thermal camera mounted on a rotating platform would reduce the need for several cameras to cover the necessary area, as well as enable a single centralized monitoring system to cover the whole region without any blind spots.

Only in the presence of at least a little light can traditional night vision cameras identify pictures. And these cameras are unable to distinguish between objects (in our example, animals) in total darkness. If a requirement for

Fig. 3 ARE (Amur Tiger Re-identification in the Wild) dataset images



multi-sensor networking arises, the suggested system may be simply expanded up using more accurate thermal cameras and Amazon cloud services. Such methods allow for the deployment of many microcontrollers as slaves that may be managed by a cloud-based master computer. All of the image processing and decision-making programming for our proposed system is executed on the Raspberry Pi itself.

3.1 Deep-CNN model

Only in the presence of at least a little light can traditional night vision cameras identify pictures. And these cameras are unable to distinguish between objects (in our example, animals) in total darkness. If a requirement for multi-sensor networking arises, the suggested system may be simply expanded up using more accurate thermal cameras and Amazon cloud services. Such methods allow for the deployment of many microcontrollers as slaves that may be managed by a cloud-based master computer. All the image processing and decision-making programming for our proposed system is executed on the Raspberry Pi itself. Many convolutional layers are produced toward the conclusion of this layer. It is challenging to review each convolutional layer separately. Here is when the Max Pooling Layer, the second layer, comes in handy. Each convolutional layer is transformed by this layer into a 2-D matrix of a lower dimension, such as 2×2 or 4×4 . Therefore, larger matrices are transformed into matrices of lower order at this layer. The third layer, known as the Flattening layer, is then used to combine this 2-D vector into a single vector. This layer takes each pooled layer individually and turns it into a one-dimensional vector. The final vector, which will be applied to match the various attributes, is this one-dimensional vector. Finally, the Dense Layer of CNN is used to format all the vectors. The model is ready to be tested once it has completed the training phase. Images of the creatures that need to be detected serve as the inputs.

The 30 training epochs were used to train this proposed CNN model. The dataset is broken up into batches and processed separately because epochs cannot be used on very big datasets. Each epoch yields a forward pass and a reverse pass of every training sample. The model's accuracy rises with each epoch. To boost the model's effectiveness, more epochs might be used.

3.2 Convolution layer

The Convolution layer, which does the majority of the calculations, is the foundational component of CNN (Malibari et al. 2022; Sangaiah et al. 2022; Patil and Ansari 2022; Panda et al. 2022; Dahou et al. 2022). A group of filters that can be learned make up the convolution layer.

Every filter has a narrow spatial range (in width and height), but it covers the whole depth of the input volume. For instance, the size of a common filter on the first layer of a convolution layer can be $3 \times 3 \times 3$. (i.e. 3 pixels width and height, and 3 color channels). By moving each filter over the input volume's width and height, we combine, and then we calculate the dot products between the input and the filter coefficient at any place. A 2-dimensional activation map, which shows the responses of that filter at each place, is constructed as the filter is moved around the input volume. To put it another way, the network will initially learn features like an edge with a certain orientation or a splotch of a particular color, and then finally a full texture or body part on higher levels of the network. The convolution layer will, intuitively, have a number of filters, each of which will generate a unique two-dimensional activation map (Sajith and Nagarajan 2020; Rajalakshmi and Minu 2014).

3.3 Pooling layer

The purpose of the pooling layer is to scale back the spatial dimension of the representation, which lowers the number of parameters and computations required by the network and also helps to prevent over-fitting. The input is resized spatially by the Pooling Layer using the MAX operation, and it does this individually for each depth slice of the input. The most typical configuration is a pooling layer with filters of size 2 administered with a stride of 2, every depth slice in the input by 2 along both width and height, and a 75% activation discard rate.

3.4 Fully connected layer

As in conventional neural networks, neurons in a fully connected layer have complete connections to every activation in the layer below. With a matrix multiplication and bias offset, their activations may be calculated. There are two critical components of CNN, Extraction of Features and Classification.

3.5 Feature extraction

The CNN network's name is derived from a feature called feature extraction, which is a component of the CNN design (Naveenkumar et al. 2022). This algorithm's effectiveness depends heavily on the mathematical action of convolution. The image we want to classify enters the red region as its input, and a set of features comes out as its output. Consider the details of an image. For instance, a picture of a tiger can show details like whiskers, two ears, four legs, etc.

In CNN, a filter or kernel is used to conduct convolution on an input picture. Beginning in the top left corner and

moving down, the filter moves over the input image one pixel at a time. The filter multiplies its values by the image's overlapping values as it slides over the image, then adds all of the importance to produce a single value for each overlap. We calculate the other output matrix values similarly. It is important to note that the output matrix's top left value, 4, depends only on the nine values (3 3) located in the top left of the original image matrix. Even if the image's other values alter, this value remains unchanged. Our CNN's output value or neuron has this as its receptive field.

The output generation of CNN using the following formula

$$y_i^l = f(\sum v^l y_i^{l-1} + b_i^l) \quad (1)$$

where v^l represents the kernels of convolutional layers and b_i^l denotes the bias of the particular layer $f(x)$ denotes the activation function.

$$f(x) = \max(0, x) \quad (2)$$

The input for the pooling layer from the convolution layer is generated as follows,

$$y_i^l = f(\beta_i^l d(y_i^{l-1}) + b_i^l) \quad (3)$$

The $d(y_i^{l-1})$ describe the down sampling of values.

Each value in our output matrix only affects a single portion of our original image. A mathematical function

known as an activation function is applied to the output after our filter has been applied to the source image. Rectified Linear Unit, or ReLU, is the activation function typically employed in most CNN feature extraction scenarios, which maintains the positive numbers identical and merely changes all negative values to 0. Adding a pooling or sub-sampling layer to CNN layers is typical to practice after obtaining the feature maps. The network's parameters and computations are reduced thanks to pooling, which also reduces the network's dimensionality. In doing so, over-fitting is prevented, and training time is reduced. Max pooling, which uses the highest value available within a specific window, is the most common. Like our previous kernel sliding procedure, the windows are comparable. By doing this, the size of the feature map is reduced while maintaining the essential details.

4 Experimental setup

4.1 Dataset

The Russian Far East and Northeast China are the central regions where the Amur tiger population is concentrated. Given that there are just 600 people left in the wild, conservation efforts are vital. To preserve natural life, it is crucial to re-identify species. More than 8000 video clips of 92 different Amur tigers from 10 Chinese zoos were

Fig. 4 Sample images of Google Open Images V6+ dataset



Table 1 Sample images from animals detection images dataset

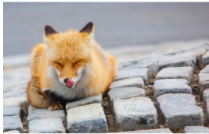
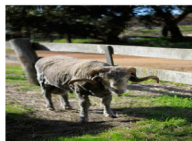
S.no	Animal	Number of data	Sample image
1	Raccoon	103	
2	Horse	400	
3	Goat	202	
4	Mule	61	
5	Duck	542	
6	Owl	406	
7	Bear	87	
8	Chicken	388	
9	Fox	148	
10	Pig	190	
11	Sheep	99	
12	Cattle	70	

Fig. 5 The proposed model setup

Model: "sequential"		
Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 126, 126, 64)	1792
dense (Dense)	(None, 126, 126, 64)	4160
dropout (Dropout)	(None, 126, 126, 64)	0
dense_1 (Dense)	(None, 126, 126, 64)	4160
dropout_1 (Dropout)	(None, 126, 126, 64)	0
flatten (Flatten)	(None, 1016064)	0
dense_2 (Dense)	(None, 128)	130056320
dense_3 (Dense)	(None, 4)	516
Total params: 130,066,948		
Trainable params: 130,066,948		
Non-trainable params: 0		

gathered by a third-party company (MakerCollider) with the assistance of WWF. They have coordinated efforts to create the ATRW (Amur Tiger Re-identification in the Wild) dataset by annotating sampled video frames with bounding-box, keypoint-based pose, and identity annotations. This dataset is utilized by us for the detection of animal intrusion. Figure 3 shows the sample images taken from the ARE (Amur Tiger Re-identification in the Wild) dataset.

Google Open Images V6+ was used to obtain the Animals (Object) Detection dataset. Animal classes have been added in this dataset are: dog, cat, zebra, lion, leopard, cheetah, tiger, bear, brown bear, butterfly, canary, crocodile, polar bear, bull, camel, crab, chicken, centipede, cattle, and caterpillar, duck. Figure 4 shows the images from Google Open Images V6+ dataset.

Table 1 shows the sample images from Animals Detection Images Dataset. For the experimental purpose we have selected 12 different animals and the corresponding data from dataset.

4.2 Proposed model setup

We employ tenfold cross-validation, where the entire feature dataset is split into ten identical folds. Tenfold cross validation is used mostly to make sure that results are still unbiased to the partitioned data being used. Nine of the ten folds are utilized for training purposes, and the tenth is used for testing purposes. As a result, 90% of the data are utilized for training and 10% are used for testing. Ten repetitions are then added to the procedure so that each sample can serve as test data. The average of all ten results constitutes the result in question. The vast array of capabilities and enhancements offered by the new Raspberry Pi 3 Model B+ will be advantageous to designers, programmers, and engineers. Pi systems are another option for engineers to incorporate into their creations. Like the other Pi computers, this raspberry Pi is equally proficient in every area. The speed of the new models must either be significantly faster than the old ones' or at least comparable. As an illustration, the CPU has been clocked at 1.4 GHz, which is 200 MHz faster than the Pi 3 B, and the Ethernet speed has been increased from 100 to 300 Mbps.

The experimental hardware settings include an Intel Core i7-8700 CPU and 16 GB of RAM. The software

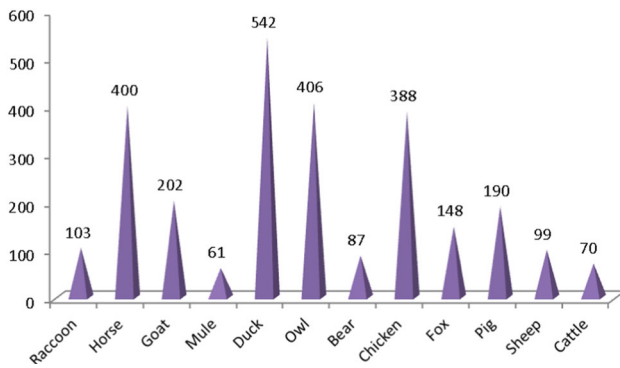


Fig. 6 Sample data distribution in animal detection dataset

environment employed in this paper includes Python 3.7, Tensor flow 2.1, and Keras2.24. The Adam optimizer and the category_cross-entropy loss function were used to train the model. Adam's learning rate is 0.0001, the epoch is 2000, the batch size is 128, the momentum in batch normalization is 0.85, and the leaky ReLu alpha is 0.2. Dropout is set to 0.4 and CNN Dropout to 0.01. Figure 5 depicts the proposed model setup.

5 Results and discussion

Our system offers a reliable and effective mechanism for detecting and analyzing animals according to the results and performances. As we have captured photographs of animals from various angles for animal-background verification, we can see that our system is resilient to poses. Additionally, because our database includes pictures from both categories, namely daytime and nocturnal images, the procedure functions well in daylight and night. The method is resistant to dynamic natural sceneries and crowded animal photographs because the database comprises camera

trap photos. Figure 6 shows the data distribution from the animal detection dataset.

With an F1 measure up to 0.95, animal detection dataset has a 92% accuracy rate on the ARE (Amur Tiger Re-identification in the Wild) dataset and 99.6% accuracy and 0.99 F1 score on the animal detection dataset and 95.6% accuracy and 0.96% of F1 score on Google Open Images V6+ dataset.

Figure 7 shows the training and validation accuracy on the ARE (Amur Tiger Re-identification in the Wild) dataset and Fig. 8 depicts the training and validation loss of ARE (Amur Tiger Re-identification in the Wild) dataset.

Figure 9 and 10 shows the accuracy and loss of the animal detection dataset.

Accuracy, recall, and precision are used to gauge the model's performance, whereas F1 score and accuracy are crucial elements in the data collecting and measurement process. Both precision and accuracy forecast the measurement in comparison with the first result. Precision, on the other hand, evaluates the reproducible value even when it deviates from the intended output while accuracy measures the value that is expected.

$$\text{Acc} = \frac{\text{Total True prediction}}{\text{Total prediction}} \quad (4)$$

$$\text{Precision} = \frac{\text{TP}}{P} \quad (5)$$

The True Positive is abbreviated as TP. TP represents only positive values. Recall is used to determine a True Positive, and the F1 score is the average of precision and recall.

$$\text{F1Score} = \frac{2 * (R * P)}{(R + P)} \quad (6)$$

where

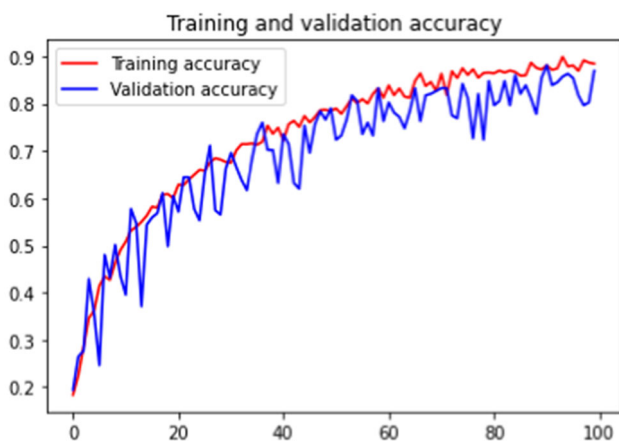


Fig. 7 Accuracy of ARE (Amur Tiger Re-identification in the Wild)

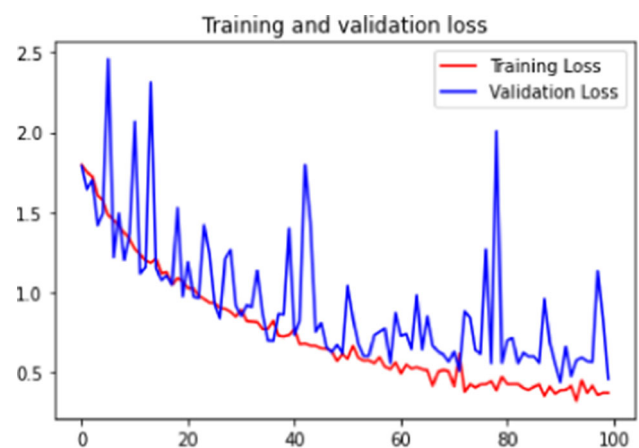


Fig. 8 Loss of ARE (Amur Tiger Re-identification in the Wild)

Fig. 9 Accuracy of animal detection dataset

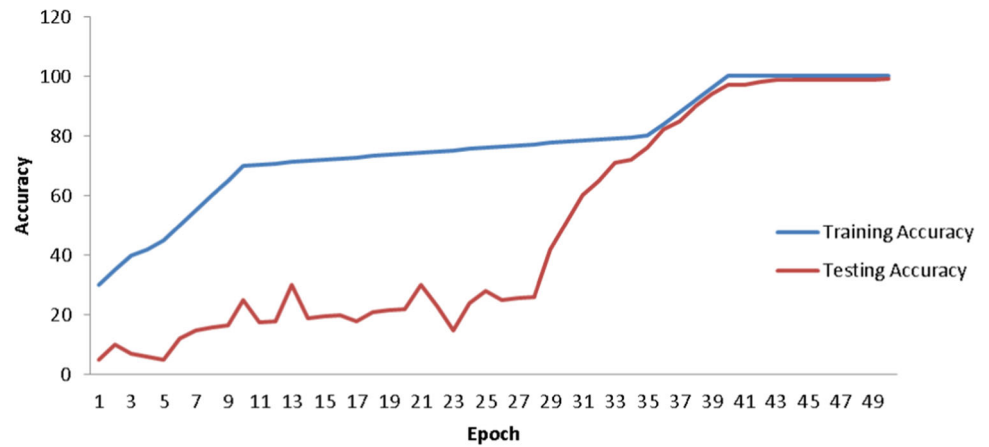


Fig. 10 Loss of animal detection dataset

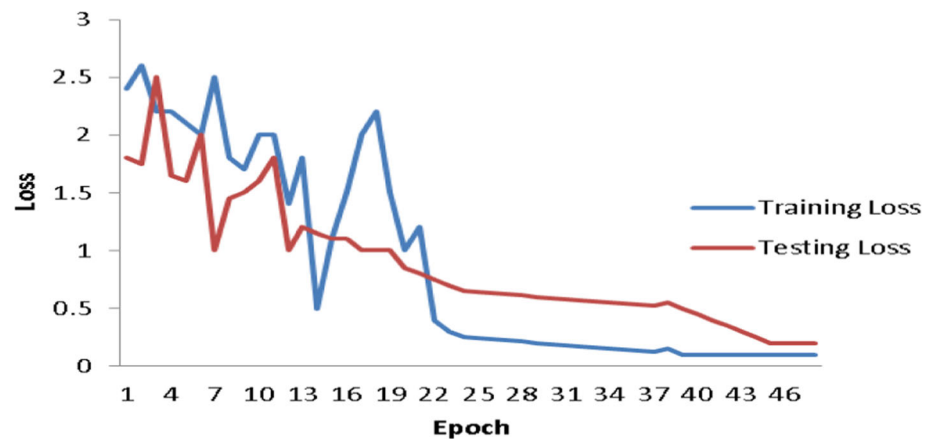
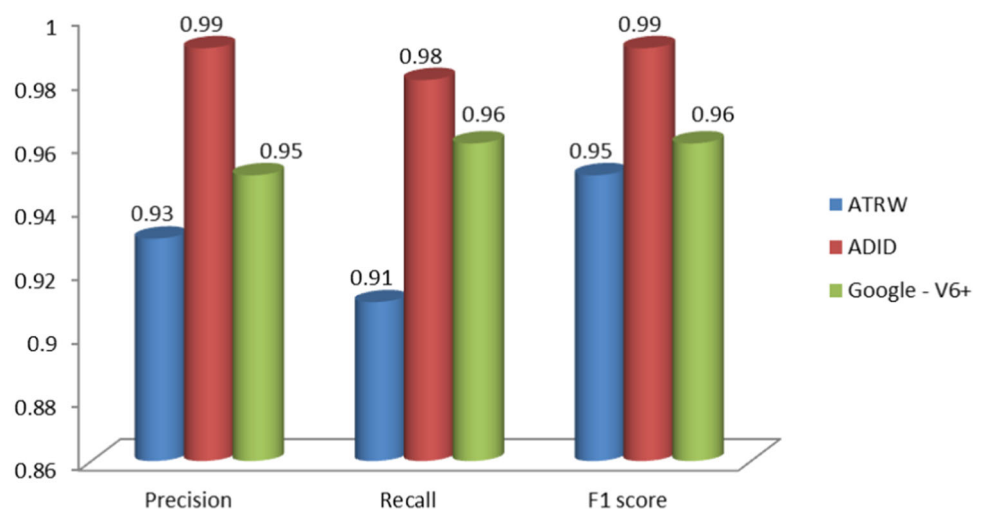


Fig. 11 Precision, recall and F1score on different dataset



$$R = \frac{\text{correctly predicted positive value}}{\text{Total positive value}}$$

(7)

R is the Recall and P is the Prediction. Table 2 shows the performance comparison on the different dataset used in our proposed method.

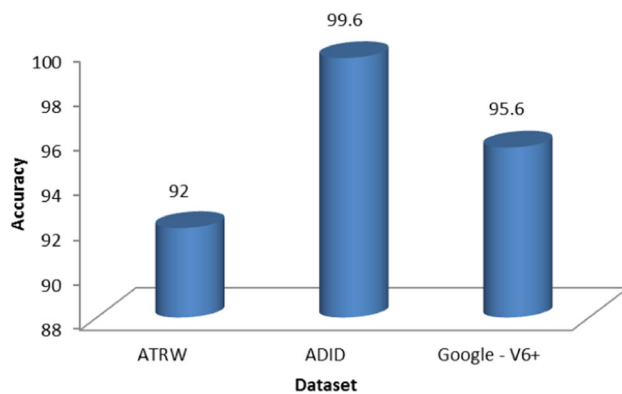


Fig. 12 Accuracy of our proposed model

Table 2 Performance comparison on the different models

Performance measures/dataset	ATRW	Animals detection images dataset	Google open images V6+
Accuracy	92	99.6	95.6
Precision	0.93	0.99	0.95
Recall	0.91	0.98	0.96
F1 score	0.95	0.99	0.96
loss	0.2542	0.2121	0.1250

Figure 11 and 12 show the precision recall and *F1* score of our proposed methodology on the dataset ATRW, Animals Detection Images Dataset and Google Open Images V6+.

6 Conclusion and future works

Animals are frequently put to death in revenge or to avoid a clash of destiny because the annoyance of wild animals damaging crops has fast become societal trouble. Many of the animals slaughtered in this way may already be threatened or endangered. It needs an engaging problem and an effective solution. As a result, since this activity is being done to address this issue, it must first be qualified as having social relevance. In order to solve this issue, we considered extending the system to include a digital camera that could display the field. As a result, the field space must be regularly watched over to keep wildlife out and to seize the intruder's photo so they can go inside. In order to create a Convolutional Neural Network and train the image dataset of a monkey, a boar, and an elephant, the proposed model will do so. The saved model will be executed on the driver code to compare the previously trained photos with the fresh test images obtained during live capture. During a

live capture, if one of the trained animals is found, speakers are used to produce a repulsive sound that will cause the animal to flee.

The approaches described in this study could be used to develop an effective system for detecting animal intrusions in places particularly vulnerable to animal attacks. By identifying the existence of animals and evaluating the hazards they pose right now, they could significantly reduce casualties caused by wild animal transgressions.

Author contributions Not applicable.

Funding The authors did not receive financial support from any organization for the submitted work.

Availability of data and material Not applicable.

Code availability Not applicable.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Consent to participate Not applicable.

Consent for publication Authors give consent to the Journal to publish their article.

References

- Angadi S, Katagall R (2019) Agrivigilance: a security system for intrusion detection in agriculture using raspberry pi and opencv. *Int J Sci Technol Res* 8(11):1260–1267
- Dahou A, Elaziz MA, Chelloug SA, Awadallah MA, Al-Betar MA, Al-qaness MAA, Forestiero A (2022) Intrusion detection system for IoT based on deep learning and modified reptile search algorithm. *Comput Intell Neurosci* 2022
- Kamencay P, Trnovszky T, Benco M, Hudec R, Sykora P, Satnik A (2016) Accurate wild animal recognition using PCA, LDA and LBPH. In: 2016 ELEKTRO, Strbske Pleso, pp 62–67
- Li Y, Huang H, Xie Q, Yao L (2018a) Research on a surface defect detection algorithm supported MobileNet-SSD. *IEEE Trans Pattern Anal* 34:743–761
- Li L, Zhang H, Peng H, Yang Y (2018b) Nearest neighbors based density peaks approach to intrusion detection. *Expert Syst Appl* 110:33–40. <https://doi.org/10.1016/j.chaos.2018.03.010>
- Liu W, Anguelov D, Erhan D, Szegedy C, Reed S, Fu CY, Berg AC (2016) Ssd: single shot multibox detector. In: Leibe B, Matas J, Sebe N, Welling M (eds) *Computer vision—ECCV 2016*. Springer, Cham, pp 21–37. <https://doi.org/10.1007/978-3-319-46448-02>
- Malibari AA, Alotaibi SS, Alshahrani R, Dhahbi S, Alabdan R, Al-wesabi FN, Hilal AM (2022) A novel metaheuristics with deep learning enabled intrusion detection system for secured smart environment. *Sustain Energy Technol Assess* 52:102312
- Naveenkumar M, Manoj R, Nandhakumar B, Rahul R (2022) DenseNet201 for Animal detection and repellent system. In:

- 2022 International conference on electronic systems and intelligent computing (ICESIC). IEEE, pp 213–218
- Panda PK, Kumar CS, Vivek BS, Balachandra M, Dargar SK (2022) Implementation of a wild animal intrusion detection model based on internet of things. In: 2022 Second international conference on artificial intelligence and smart energy (ICAIS). IEEE, pp 1256–1261
- Patil HD, Ansari NF (2022) Intrusion detection and repellent system for wild animals using artificial intelligence of things. In: 2022 International conference on computing, communication and power technology (IC3P). IEEE, pp 291–296
- Rajalakshmi T, Minu RI (2014) Improving relevance feedback for content based medical image retrieval. In: International conference on information communication and embedded systems (ICICES2014). IEEE, pp 1–5
- Sailesh K, Balina HV, Sivakumar T, VijayaPoojitha P (2019) Detection of wild elephants using image processing on raspberry pi3. *Int J Comput Sci Mob Comput* 8(2):104–115
- Sajith PJ, Nagarajan G (2021) Optimized intrusion detection system using computational intelligent algorithm. In: *Advances in electronics, communication and computing: select proceedings of ETAEEERE 2020*. Springer, Singapore, pp 633–639
- Sajith PJ, Nagarajan G (2022) Network intrusion detection system using ANFIS classifier. *Soft Comput* 1–10
- Sangaiah AK, Javadpour A, Ja'fari F, Pinto P, Zhang W, Balasubramanian S (2022) A hybrid heuristics artificial intelligence feature selection for intrusion detection classifiers in cloud of things. *Cluster Comput* 1–14
- Thyagarajan KK, Minu RI (2013) Prevalent color extraction and indexing. *Int J Eng Technol* 5(6):4841–4849
- Trnovszky T, Kamencay P, Orješek R, Benco M, Sykora P (2017) Animal recognition system based on convolutional neural network. *Adv Electr Electron Eng*. <https://doi.org/10.15598/aeee.v15i3.2202>
- Vinayakumar R, Alazab M, Soman KP, Poornachandran P, Al-Nemrat A, Venkatraman S (2019) Deep learning approach for intelligent intrusion detection system. *IEEE Access* 7:41525–41550. <https://doi.org/10.1109/ACCESS.2019.2895334>
- Zhang H, Huang L, Wu CQ, Li Z (2020) An effective convolutional neural network based on SMOTE and Gaussian mixture model for intrusion detection in imbalanced dataset. *Comput Netw* 177:107315

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